Continuous Random Variable:

A random variable X is said to be continuous, if its Range is an uncountable set.

Example: Let X be a random variable that denotes the number that is chosen uniformly at random from $[0,1]$.

Then clearly the $\text{Range}(X)$ is $[0,1]$, which is an uncountable set.

Choosing a number uniformly at random means,

$$\Pr(X \in [x_1, x_2]) \propto x_2 - x_1$$

where, $0 \leqslant x_1 \leqslant x_2 \leqslant 1$

For a continuous random variable X,

$$\Pr(X = x) = 0 \quad , \text{ where } x \in \mathbb{R}.$$

Uniform Distribution.

Let a, b be any two real numbers such that $a \leqslant b$. Suppose we choose a number uniformly at random from $[a,b]$. Let X be a random variable that denotes the chosen number. Then it clearly makes no sense to ask what is the probability that $X = \frac{a+b}{2}$, because it is 0.

But clearly, $Pr(X \in [a,b]) = 1$

So,

Probability across $b-a$ length is ———— 1

Then

Probability across a unit length is ———— $\frac{1}{b-a}$

Finally,

Probability across $x_2 - x_1$ length is ———— $\frac{x_2 - x_1}{b-a}$

Probability Density Function (PDF):

Informally, PDF tells us the probability per unit.

for uniform distribution, as we saw above, probability per unit length is

$$\frac{1}{b-a}.$$

formally, for a continuous random variable X, PDF is a function, usually denoted by f,

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

such that,

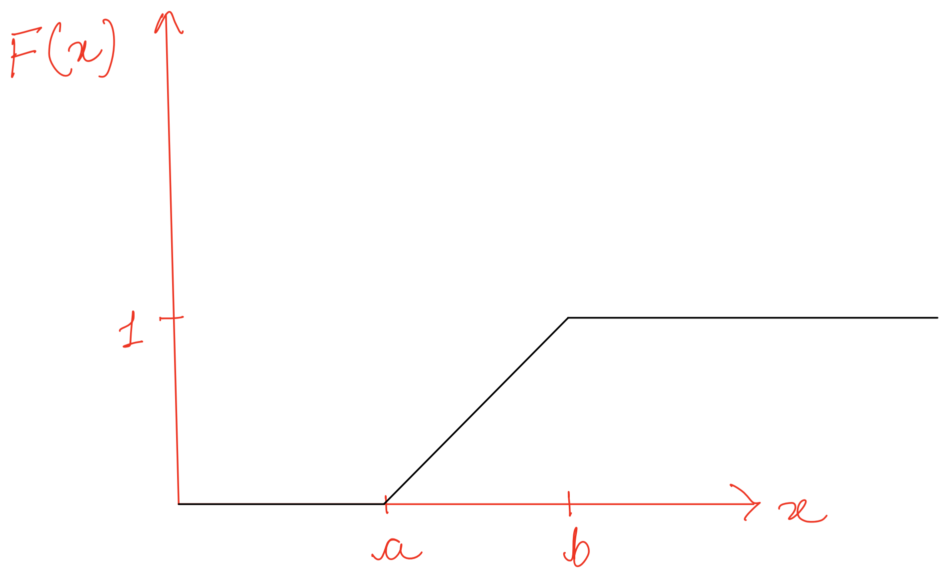
$$f(x) = \frac{d(F(x))}{dx}$$

where, F is CDF of x.

CDF of a Uniformly Distributed Random Variable :

If X is a uniformly distributed random variable, then

$$F(x) = \begin{cases} 0 & , \text{if } x < a \\ \dfrac{x-a}{b-a} & , \text{if } a \leq x \leq b \\ 1 & , \text{if } x > b \end{cases}$$



Expected Value of a Continuous Random Variable:

For a continuous random variable X,

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\,dx$$

Variance of a Continuous Random Variable:

For a continuous random variable X,

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

where $\mu = E[X]$.

Expectation and Variance of a Uniformly distributed random variable:

If $X \sim \text{Uniform}(a,b)$

then,

$$E[X] = \frac{a+b}{2}$$

and

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

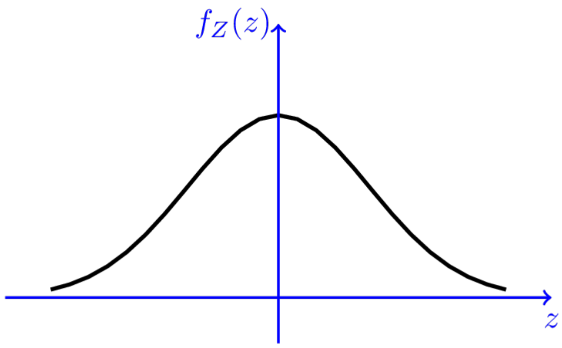
Normal Distribution :

A continuous random variable is said to be a standard normal (Gaussian) random variable, if its PDF is given as,

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-z^2/2\right) , \text{ for } z \in \mathbb{R}.$$

$$\exp\left(-z^2/2\right) = e^{\left(-z^2/2\right)}.$$

A standard Gaussian random variable is denoted by, $N(0,1)$.



PDF of $Z \sim N(0,1)$

Image Credit : https://www.probabilitycourse.com/chapter4/4_2_3_normal.php

Expectation and Variance of a standard Gaussian random variable Z:

$$E[Z] = 0$$

$$Var(Z) = 1$$

CDF of the standard normal distribu- is more generally denoted as, Φ.

$$\Phi(x) = P_r(Z \leqslant x)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{x} \exp\left(\frac{-u^2}{2}\right) du$$

Normal random variable (function of a standard normal random variable)!

If Z is a standard normal random variable, i.e., $Z \sim N(0,1)$, then

$$X = \sigma Z + \mu \quad , \text{ where } \sigma > 0$$

is a normal random variable with mean μ and variance σ^2.

So, $X \sim N(\mu, \sigma^2)$.

If $X \sim N(\mu, \sigma^2)$, then CDF of X is,

$$F(x) = \Pr(X \leq x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

$$\Pr(a < X \leq b) = \Pr(X \leq b) - \Pr(X \leq a)$$
$$= \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$